Fine-Tuning Whisper for Informal Spanish Speech Recognition

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Abstract

We fine-tune OpenAI Whisper-small on VoxPopuli and Multilingual LibriSpeech Spanish and benchmark it against zero-shot Whisper variants and two commercial APIs.

Fine-tuning reduces VoxPopuli WER by 48% relative (18.8% to 9.7%), approaching ElevenLabs Scribe (8.7%).

On personal informal recordings from three native speakers, speaker-adaptive fine-tuning achieves near-zero WER (0.2%, 0.1%, 2.0%), outperforming all commercial APIs with minimal speaker data

Introduction

Large-scale speech recognition models such as Whisper have demonstrated strong zero-shot performance across dozens of languages, including Spanish, by training on hundreds of thousands of hours of internet-collected audio. However, this pre-training data is heavily skewed toward formal speech domains — parliamentary proceedings, audiobook narration, and broadcast news — which are over-represented on the web.

Informal conversational speech presents a substantially different challenge: speakers vary in accent and rhythm, recordings are made on personal devices under uncontrolled acoustic conditions, and sentences lack the structured formality of read text. These factors combine to produce systematic WER degradation, even in models as capable as Whisper.

This work asks a practical question: can a modest fine-tuning procedure on publicly available Spanish corpora close this gap for informal personal recordings, and how does the resulting open-source model compare against state-of-the-art commercial ASR APIs? To answer this, we evaluate six systems — three zero-shot Whisper variants of increasing size, a fine-tuned Whisper-small trained on VoxPopuli and Multilingual LibriSpeech Spanish, and two commercial APIs (Google Chirp and ElevenLabs Scribe) — across both standard benchmarks and a personal recording set collected from three native Spanish speakers under everyday conditions.

Methods and Materials

Table 1. Training data.

SOURCE	SPLIT USED	SAMPLES
VoxPopuli ES	train	5000
MLS Spanish	train	5000
Personal recordings	train	124

Model: openai/whisper-small — 244M parameters, encoder-decoder transformer (12 encoder + 12 decoder layers, d_model = 768).

Table 2. Training configuration.

Hyperparameter	Value
Optimizer	AdamW
Learning Rate	1×10^{-5} (linear decay)
Warmup steps	500
Total steps	4000
Effective Batch Size	16
Mixed Precision	BF16
Hardware	NVIDIA RTX 5060 Laptop 8 GB
Best checkpoint	Step 2,000 (val WER 10.51%)

Evaluation: 200 samples each from VoxPopuli ES test and MLS Spanish test. All personal recordings per speaker. Metrics: WER and CER via jiwer, normalized text (lowercase, punctuation removed).

Results

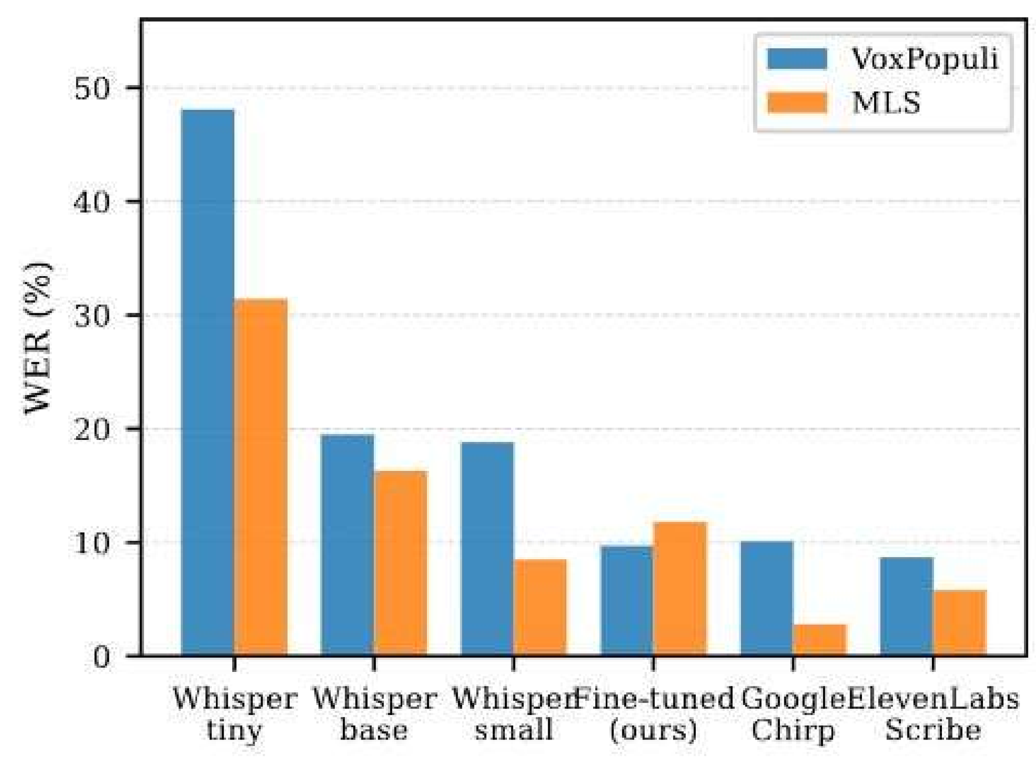
Table 3. Benchmark WER (%) — 200 samples local models, †100 samples APIs.

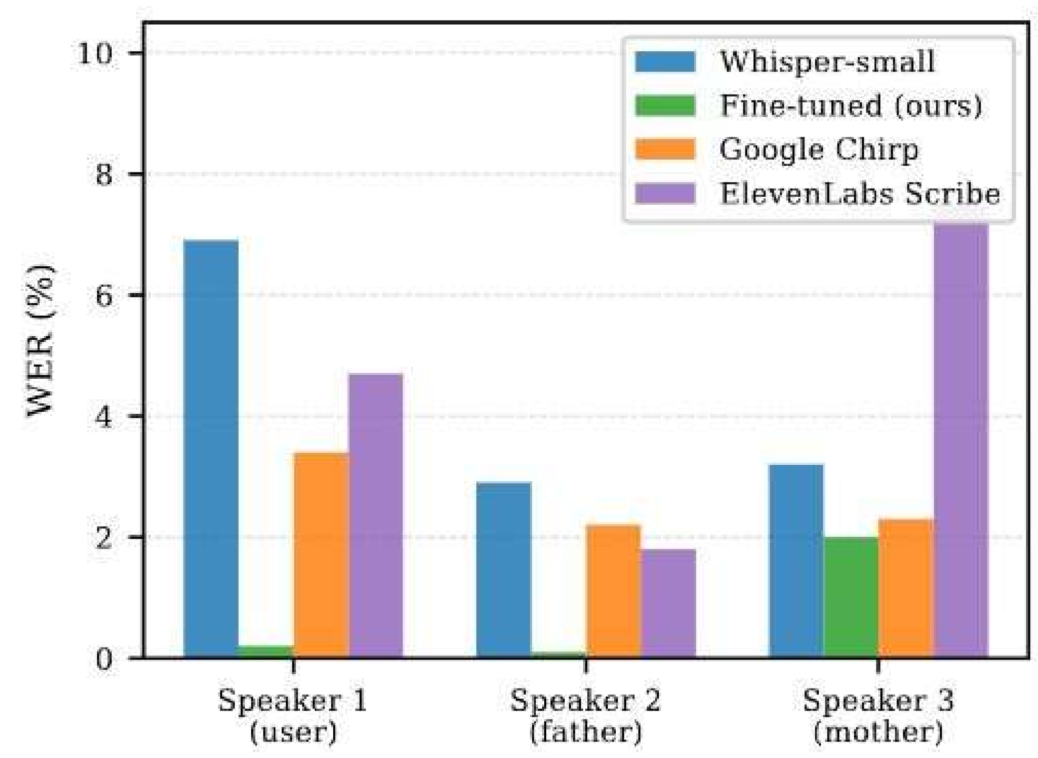
Model	VoxPopuli	MLS
Whisper-tiny (zero-shot)	48.1	31.4
Whisper-base (zero-shot)	19.5	16.3
Whisper-small (zero-shot)	18.8	8.5
Whisper-small (fine-tuned)	**9.7**	**11.8**
Google Chirp†	10.1	2.9
ElevenLabs Scribe†	8.7	5.8

Table 4. Personal recordings WER (%) — speaker adaptation

Model	Speaker 1	Speaker 2	Speaker 3
Whisper small (zero-shot)	6.9	2.9	3.2
Whisper small (fine-tuned)	**0.2**	**0.1**	**2.0**
Google Chirp†	3.4	2.2	2.3
ElevenLabs Scribe†	4.7	1.8	7.5‡

‡ one network timeout, result may be inflated





Discussion

- Fine-tuning yields −48% relative WER on VoxPopuli (18.8% to 9.7%), closing most of the gap to ElevenLabs Scribe (8.7%).
- MLS regression (8.5% to 11.8%) is expected: the training corpus is weighted toward parliamentary speech, shifting the acoustic prior away from clean read audiobook narration.
- Zero-shot Whisper-small already handles informal Spanish well (2.9–6.9% on personal audio); commercial APIs reach 1.8–4.7% without any speaker data.
- Speaker-adaptive fine-tuning with ~30 minutes of personal data reduces WER to near zero (0.2%, 0.1%, 2.0%), outperforming every commercial API on these speakers.
- Google Chirp is the strongest commercial reference on unseen informal speakers (2.2–3.4%); ElevenLabs Scribe is strongest on spontaneous benchmark speech (8.7% VoxPopuli).

Conclusions

Fine-tuning Whisper-small on 10,000 samples reduces VoxPopuli WER by 48% relative to the zero-shot baseline, matching commercial API performance on spontaneous speech.

A learning rate of 10^{-5} with warmup effectively adapts a 244M-parameter model without catastrophic forgetting.

The fine-tuned model is deployed in a real-time FastAPI + JavaScript application with Claude-powered language coaching feedback.

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